Works well in challenging environments

The Gazelle platform supports diagnostics for both malaria and sickle cell in an easy-to-transport, battery-powered device consisting of disposable cartridges and one reader. It can withstand extreme temperatures and humidity.

The Power of the Laboratory
At the Point of Need

gazelle®

A novel malaria and sickle cell diagnostic platform

Rapid, Accurate Performance

Gazelle identifies malaria more quickly and accurately than today's malaria diagnostics when compared with standard screening tests.

	Time to Results	Cost Per Test	Training
Microscopy	45 min	Medium	High
RDTs	20 min	Low	Medium
Hemex	1 min	Low	Low

N=85 Primarily P. vivax	Gazelle	RDT (SD Bioline)	Microscopy
Sensitivity	95.9	87.8	98.0
Specificity	97.2	100	100
Accuracy	96.5	92.9	98.8

Table 1. Clinical Study in Iquitos, Peru, using PET-PCR as the gold standard

Presented to Military Health System Research Symposium, 8/2019

(Intermediate results; study still in progress)

N=262 Primarily P.falciparum	Gazelle	RDT (SD Bioline)
Sensitivity	97.6	100
Specificity	96.8	92.3
Accuracy	96.9	93.5

Table 2. Clinical Study in Jabalpur, Madhya Pradesh, India, using microscopy as the gold standard.

Kumar R, Verma AK, Shrivas S, et al. First successful field evaluation of new, one-minute hemozoin-based malaria diagnostic device. *EClinicalMedicine* 2020; 22: 100347.

Gazelle® showed high accuracy, sensitivity, and specificity in detecting SCD (HbSS), Sickle Cell Trait (HbAS), HbE Trait when compared with standard screening tests.

Category N=726	Hemoglobin Type	Accuracy
Sickle Cell Disease	HbSS	100%
Sickle Cell & HbE Trait	HbAS, HbAE	97.6%
Normal	HbAA	97.8%
All Categories	-	97.9%

Table 3: Clinical study in Madhya Pradesh and Chhattisgarh in India using HPLC/electrophoresis as gold standard.

Presented to American Society of Tropical Medicine and Hygiene, 11/2019

GZL-MKG-0003-1 r1

Mobile Health

The diagnostic reader can store or upload patient data to a USB drive for later storage in the cloud.* The GPS location, useful for epidemiological studies, is also saved.

* Refer to https://www.hemexhealth.com for latest updates

A Solution for Today and the Future

Gazelle is designed to deliver value at an affordable price.

Gazelle technologies will be further developed to detect more diseases using the same reader. Hemex's novel diagnostics can help fuel the worldwide trend toward more patient-centered care.

An Experienced Team

Our product combines technological invention from university scientists at Case Western Reserve University (CWRU) with innovation from entrepreneurs who have launched over 60 products.

After years of introducing successful products into the developing world, the founders have established a strong network of contacts within key markets. Hemex is building relationships with corporate partners, international NGOs and key government organizations, as well as clinical experts.

For more information, please contact:

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or go to https://www.hemexhealth.com

Gazelle has CE Mark for Hb Variant and Malaria, but is not yet approved in all countries. Please check website for the latest update.

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Hemex Health

Creating affordable, life-changing medical diagnostics for under-served people everywhere

Malaria & Sickle Cell

Accurate and easy-to-use test finds malaria in about 1 minute

Early and accurate diagnosis is important in the fight against malaria. Microscopy is the gold standard for most areas, but the process is time consuming and skilled microscopists with proper equipment are often not available, especially in remote areas.

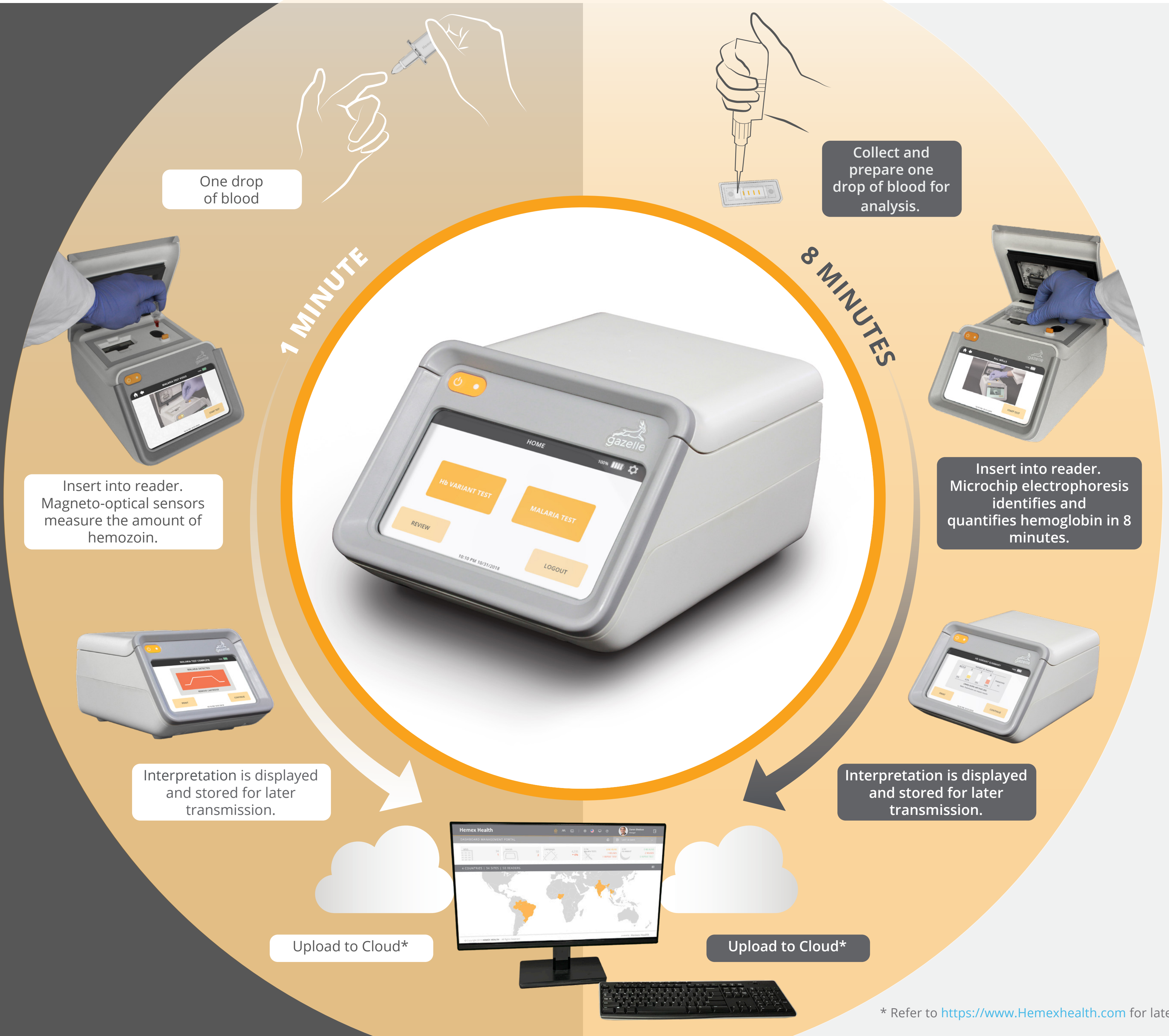
Improved malaria diagnostics are needed that are not only cost-effective but also easy to use, rapid, and highly accurate.

Gazelle's speed and accuracy make it the ideal for screening large populations during epidemics or at border crossings.

Gazelle is a more sensitive test for *P. vivax* than RDTs

Gazelle uniquely identifies hemozoin (a byproduct of all species and variants of malaria parasites) even at low levels. The iron-bearing hemozoin are manipulated and measured by Gazelle's magneto-optical sensors to detect the presence of malaria. Since hemozoin is present in all species, results are not impacted by HRP2 gene deletion, which can affect the performance of RDTs. In clinical testing, Gazelle has detected malaria from *P. falciparum*, *P. vivax*, *P. ovale,* and *P. malariae* species.

In both field and dilution studies, Gazelle significantly outperformed RDTs in detecting low level infections of *P. vivax* malaria.



Accurate interpretation for sickle cell disease or trait in 8 minutes

The highest prevalence of sickle cell occurs in areas with the least access to accurate screening tools.

Most carriers of sickle cell and other hemoglobin disorders are unaware that they carry the trait, which greatly increases the chances that they will have a child with the disease. Hundreds of children die needlessly each day from complications arising from sickle cell because they were never diagnosed.

Gazelle offers affordable, accurate testing. Sample analysis is completed in 8 minutes, allowing patients to learn their status at the time of visit. Babies can be screened at the time of vaccination and immediately started on cost-effective care to live longer, healthier lives.

Miniaturized, advanced electrophoresis with AI and automated interpretation

To help clinicians diagnose sickle cell disease, Hemex invented "microchip electrophoresis", a process that separates hemoglobin types according to their charge. Gazelle performs comparably to a laboratory electrophoresis test (the gold standard), and can identify and provide percentages of hemoglobin types A2/C/E, A, S, and F (fetal). Quantification is helpful information to understand the impact of treatment and to identify other variants.

Gazelle's automated interpretation makes the test usable by healthcare workers of all levels of experience.

* Refer to https://www.Hemexhealth.com for latest updates

1,000,000,000	**400,000**	**35**	**240,000,000**	**50–80%**	**70%**
Tests are needed every year	People die every year	Countries have elimination goals by 2030	Tests needed each year for underserved regions	Of children die before their 5th birthday	Could be saved with early diagnosis